

Broadening Horizons: Exploring the Cathepsin Family as Therapeutic Targets for Alzheimer's Disease.

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Alzheimer's disease (AD) is an intricate neurodegenerative disorder characterized by the accumulation of misfolded proteins, including beta-amyloid (A β) and tau, leading to cognitive decline. Despite decades of research, the precise mechanisms underlying its onset and progression remain elusive. Cathepsins are a family of lysosomal enzymes that play vital roles in cellular processes, including protein degradation and regulation of immune responses. Emerging evidence suggests that cathepsins may be involved in AD pathogenesis. Cathepsins can influence the activation of microglia and astrocytes, the resident immune cells in the brain. However, cathepsin dysfunction may lead to the accumulation of misfolded proteins, notably A β and tau. In addition, dysregulated cathepsin activity may induce an exaggerated immune response, promoting chronic inflammation and neuronal dysfunction in patients with AD. By unraveling the classification, functions, and roles of cathepsins in AD's pathogenesis, this review sheds light on their intricate involvement in this devastating disease. Targeting cathepsin activity could be a promising and novel approach for mitigating the pathological processes that contribute to AD, providing new avenues for its treatment and prevention.

Ferroptosis in Parkinson's disease: Molecular mechanisms and therapeutic potential.

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Parkinson's Disease (PD) is characterized by the progressive loss of dopaminergic neurons in the substantia nigra (SN), leading to motor and non-motor symptoms. While the exact mechanisms remain complex and multifaceted, several molecular pathways have been implicated in PD pathology, including accumulation of misfolded proteins, impaired mitochondrial function, oxidative stress, inflammation, elevated iron levels, etc. Overall, PD's molecular mechanisms involve a complex interplay between genetic, environmental, and cellular factors that disrupt cellular homeostasis, and ultimately lead to the degeneration of dopaminergic neurons. Recently, emerging evidence highlights ferroptosis, an iron-dependent non-apoptotic cell death process, as a pivotal player in the advancement of PD. Notably, oligomeric α -synuclein (α -syn) generates reactive oxygen species (ROS) and lipid peroxides within cellular membranes, potentially triggering ferroptosis. The loss of dopamine, a hallmark of PD, could predispose neurons to ferroptotic vulnerability. This unique form of cell demise unveils fresh insights into PD pathogenesis, necessitating an exploration of the molecular intricacies connecting ferroptosis and PD progression. In this review, the molecular and regulatory mechanisms of ferroptosis and their connection with the pathological processes of PD have been systematically summarized. Furthermore, the features of ferroptosis in PD animal models and clinical trials targeting ferroptosis as a therapeutic approach in PD patients' management are scrutinized.

The GRP78-PERK axis contributes to memory and synaptic impairments in Huntington's disease R6/1 mice.

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Increasing evidence indicates that a key factor in neurodegenerative diseases is the activation of the

unfolded protein response (UPR) caused by an accumulation of misfolded proteins in the endoplasmic reticulum (ER stress). Particularly, in Huntington's disease (HD) mutant huntingtin (mHtt) toxicity involves disruption of the ER-associated degradation pathway and loss of the ER protein homeostasis leading to neuronal dysfunction and degeneration. Besides the role of the UPR in regulating cell survival and death, studies that demonstrate the contribution of sustained UPR activation, particularly of PERK signaling, in memory disturbances and synaptic plasticity deficiencies are emerging. Given the contribution of hippocampal dysfunction to emotional and cognitive deficits seen in HD, we have analyzed the involvement of ER stress in HD memory alterations. We have demonstrated that at early disease stages, ER stress activation manifested as an increase in GRP78 and CHOP is observed in the hippocampus of R6/1 mice. Genetic reduction of GRP78 expression resulted in preventing hippocampal-dependent memory alterations but no motor deficits. Accordingly, hippocampal neuropathology namely, dendritic spine loss and accumulation of mHtt aggregates was ameliorated by GRP78 reduction. To elucidate the signaling pathways, we found that the inactivation of PERK by GSK2606414 restored spatial and recognition memories in R6/1 mice and rescued dendritic spine density in CA1 pyramidal neurons and protein levels of some specific immediate early genes. Our study unveils the critical role of the GRP78/PERK axis in memory impairment in HD mice and suggests the modulation of PERK activation as a novel therapeutic target for HD intervention.

Most mutations that cause spinocerebellar ataxia autosomal recessive type 16 (SCAR16) destabilize the protein quality-control E3 ligase CHIP.

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The accumulation of misfolded proteins promotes protein aggregation and neuronal death in many neurodegenerative diseases. To counteract misfolded protein accumulation, neurons have pathways that recognize and refold or degrade aggregation-prone proteins. One U-box-containing E3 ligase, C terminus of Hsc70-interacting protein (CHIP), plays a key role in this process, targeting misfolded proteins for proteasomal degradation. CHIP plays a protective role in mouse models of neurodegenerative disease, and in humans, mutations in CHIP cause spinocerebellar ataxia autosomal recessive type 16 (SCAR16), a fatal neurodegenerative disease characterized by truncal and limb ataxia that results in gait instability. Here, we systematically analyzed CHIP mutations that cause SCAR16 and found that most SCAR16 mutations destabilize CHIP. This destabilization caused mutation-specific defects in CHIP activity, including increased formation of soluble oligomers, decreased interactions with chaperones, diminished substrate ubiquitination, and reduced steady-state levels in cells. Consistent with decreased CHIP stability promoting its dysfunction in SCAR16, most mutant proteins recovered activity when the assays were performed below the mutants' melting temperature. Together, our results have uncovered the molecular basis of genetic defects in CHIP function that cause SCAR16. Our insights suggest that compounds that improve the thermostability of genetic CHIP variants may be beneficial for treating patients with SCAR16.

Protein misfolding and neurodegeneration.

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A key molecular pathway implicated in diverse neurodegenerative diseases is the misfolding,

aggregation, and accumulation of proteins in the brain. Compelling evidence strongly supports the hypothesis that accumulation of misfolded proteins leads to synaptic dysfunction, neuronal apoptosis, brain damage, and disease. However, the mechanism by which protein misfolding and aggregation trigger neurodegeneration and the identity of the neurotoxic structure is still unclear. The aim of this article is to review the literature around the molecular mechanism and role of misfolded protein aggregates in neurodegeneration and the potential for the misfolding process to lead to a transmissible form of disease by a prion-based model of propagation.

Nanoimaging for protein misfolding diseases.

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Misfolding and aggregation of proteins are widespread phenomena leading to the development of numerous neurodegenerative disorders such as Parkinson's, Alzheimer's, and Huntington's diseases. Each of these diseases is linked to structural misfolding and aggregation of a particular protein. The aggregated forms of the protein induce the development of a particular disease at all levels, leading to neuronal dysfunction and loss. Because protein refolding is frequently accompanied by transient association of partially folded intermediates, the propensity to aggregate is considered a general characteristic of the majority of proteins. X-ray crystallography, nuclear magnetic resonance, electron microscopy, and atomic force microscopy have provided important information on the structure of aggregates. However, fundamental questions, such as why the misfolded conformation of the protein is formed, and why this state is important for self-assembly, remain unanswered. Although it is well known that the same protein under pathological conditions can lead to the formation of aggregates with diverse biological consequences, the conditions leading to misfolding and the formation of the disease prone complexes are unclear, complicating any development of efficient prevention of the diseases. Misfolded states exist transiently, so answering these questions requires the use of novel approaches and methods. Progress has been made during the past few years, when recently developed ensemble methods and single-molecule biophysics techniques were applied to the problem of the protein misfolding. In this review, the impacts of these studies on the understanding of the mechanisms of the protein self-assembly into aggregates and on the development of treatments of the diseases are discussed.

Aggregopathy in neurodegenerative diseases: mechanisms and therapeutic implication.

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Many neurodegenerative diseases such as Parkinson's, Alzheimer's, Huntington's and Lou Gehrig's disease are associated with the misfolding and aggregation of proteins. While the relevance of these aggregates for neuronal degeneration and their impact on cellular function is still a matter of debate, several experimental therapeutic approaches have been aimed at interfering with protein aggregation. In this review, we want to summarize the current understanding of aggregate formation and toxicity in neurodegenerative diseases with an emphasis on Parkinson's disease. Furthermore, we will discuss current treatment strategies in these diseases targeting aggregate formation and concurrent neuronal cell death in these diseases.

Heat shock transcription factor 1 as a therapeutic target in neurodegenerative diseases.

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Neurodegenerative diseases such as Alzheimer's disease, Parkinson's disease, Huntington's disease, amyotrophic lateral sclerosis and prion-based neurodegeneration are associated with the accumulation of misfolded proteins, resulting in neuronal dysfunction and cell death. However, current treatments for these diseases predominantly address disease symptoms, rather than the underlying protein misfolding and cell death, and are not able to halt or reverse the degenerative process. Studies in cell culture, fruitfly, worm and mouse models of protein misfolding-based neurodegenerative diseases indicate that enhancing the protein-folding capacity of cells, via elevated expression of chaperone proteins, has therapeutic potential. Here, we review advances in strategies to harness the power of the natural cellular protein-folding machinery through pharmacological activation of heat shock transcription factor 1--the master activator of chaperone protein gene expression--to treat neurodegenerative diseases.

Targeting oligomers in neurodegenerative disorders: lessons from α -synuclein, tau, and amyloid- β peptide.

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Neurodegenerative disorders such as Alzheimer's disease, Parkinson's disease, Prion disease, Huntington's disease, and amyotrophic lateral sclerosis are increasingly being realized to have common cellular and molecular mechanisms including protein aggregation and inclusion body formation in selected brain regions. The aggregates usually consist of insoluble fibrillar aggregates containing misfolded protein with β -sheet conformation. The most probable explanation is that inclusions and the aggregates symbolize an end stage of a molecular cascade of several events, and that earlier event in the cascade may be more directly tied up to pathogenesis than the inclusions themselves. Small intermediates termed as 'soluble oligomers' in the aggregation process might influence synaptic dysfunction, whereas large, insoluble deposits might function as reservoir of the bioactive oligomers. Compelling evidence suggests the role of misfolded proteins in the form of oligomers might lead to synaptic dysfunction, neuronal apoptosis and brain damage. However, the mechanism by which oligomers trigger neurodegeneration still remains mysterious. The aim of this article is to review the literature around the molecular mechanism and role of oligomers in neurodegeneration and leading approaches toward rational therapeutics.

Proteasome inhibition induces stress kinase dependent transport deficits--implications for Alzheimer's disease.

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Alzheimer's disease (AD) is characterized by accumulation of two misfolded and aggregated proteins, β -amyloid and hyperphosphorylated tau. Both cellular systems responsible for clearance of misfolded and aggregated proteins, the lysosomal and the proteasomal, have been shown to be malfunctioning in the aged brain and more so in patients with neurodegenerative diseases, including AD. This malfunction could be contributing to β -amyloid and tau accumulation, eventually aggregating in plaques and tangles. We have investigated the impact of decreased proteasome activity on tau phosphorylation as well as on

microtubule stability and transport. To do this, we used our recently developed neuronal model where human SH-SY5Y cells obtain neuronal morphology and function through differentiation. We found that exposure to low doses of the proteasome inhibitor MG-115 caused tau phosphorylation, microtubule destabilization and disturbed neuritic transport. Furthermore, reduced proteasome activity activated several proteins implicated in tau phosphorylation and AD pathology, including c-Jun N-terminal kinase, c-Jun and extracellular signal-regulated protein kinase (ERK) 1/2. Restoration of the microtubule transport was achieved by inhibiting ERK 1/2 activation, and simultaneous inhibition of both ERK 1/2 and c-Jun reversed the proteasome inhibition-induced tau phosphorylation. Taken together, this study suggests that a decrease in proteasome activity can, through activation of c-Jun and ERK 1/2, result in several events related to neurodegenerative diseases. Restoration of proteasome activity or modulation of ERK 1/2 and c-Jun function can open new treatment possibilities against neurodegenerative diseases such as AD.
